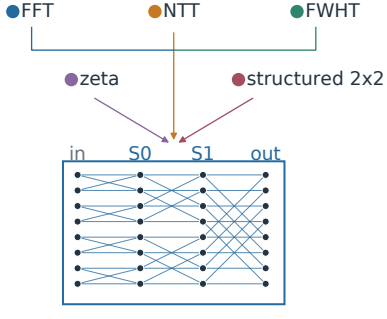
(a) Shared dependency construction

operator labels stay explicit

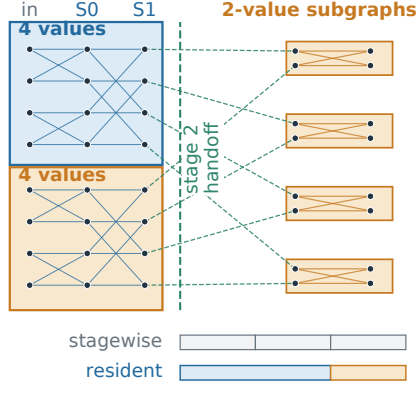


same dependency rule

operator-specific order and scaling

(b) Stage and data organization

closed bands with explicit fan-in



(c) Hierarchical physical mapping

storage and synchronization scopes

